SERVICE BULLETIN

MITSUBISHI MOTORS

AFTERSALES SERVICE OFFICE, MITSUBISHI MOTORS AUSTRALIA LTD

PURPOSE: INFORMATION	**ISSUE NO.:** 33-2010-001	**DATE:** 16-04-2010
SUBJECT: INSPECTION METHOD FOR SHOCK ABSORBER OIL LEAK		**<MODEL>** **<M/Y>** All vehicles
GROUP: 33 - FRONT SUSPENSION / 34 - REAR SUSPENSION		

1. Description:

When a shock absorber is judged to have an oil leak it is logical to replace it.

However, it has been found that a large number of serviceable shock absorbers are being mistakenly replaced under warranty.

A very small amount of oil adhered to the shock absorber doesn't immediately suggest an oil leak, but rather, it may be a normal phenomenon resulting from the functional nature of the shock absorber.

Misjudgment and the subsequent replacement of shock absorbers results in unnecessary cost.

This Service Bulletin provides clearer criteria for judging shock absorber oil leaks.

2. Applicable Manuals:

MANUAL	ATTACHMENT
All manuals containing: Group 33 - Front Suspension Group 34 - Rear suspension	Attachment 1

3. Effective Models:

All vehicles.

4. Details:

Refer to attachment 1.

5. Warranty Policy:

Warranty claims based on inadequate judgment will not be approved.
It is strongly requested that a thorough inspection be performed before submitting a warranty claim.

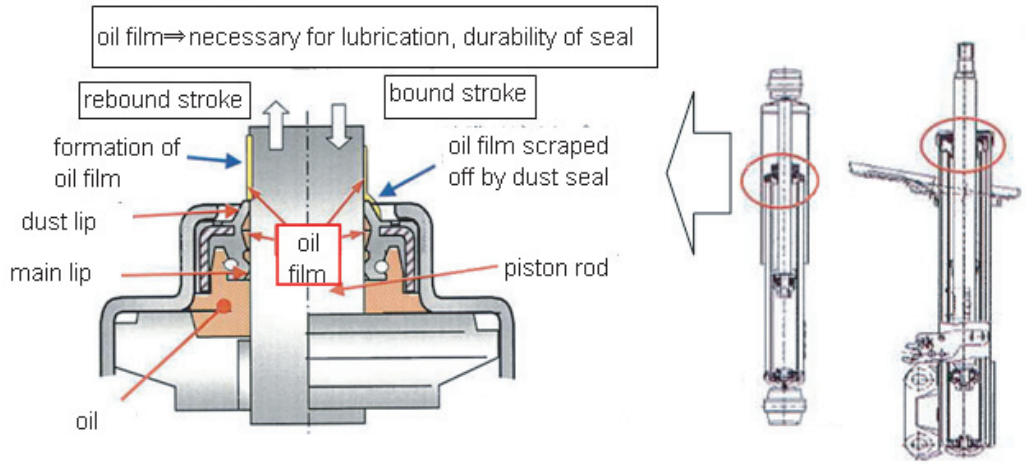
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Attachment 1 (1/2)

4. Details:

Phenonenon and Cause: Sealing structure of a shock absorber.

The sealing of shock absorber has 3 functions:
- to seal the shock absorber oil inside the cylinder,
- to prevent dust from entering into the inside structure,
- to lubricate the piston rod by forming an oil film.

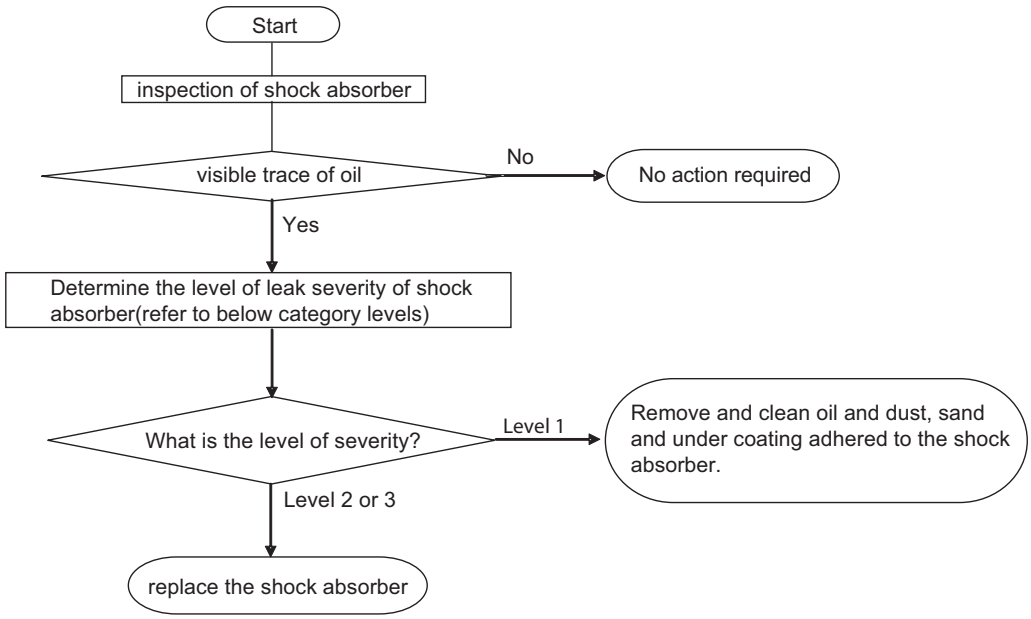


The lubricating function of the seal, as shown in the above drawing, lubricates the piston rod by its sliding motion. As a result of repetitive sliding of the piston rod, oil film scraped off the piston rod or condensed oil vapor may adhere to the surface of the shock absorber casing. This small amount of oil seen on the surface of piston rod may be recognized as an oil leak caused by defective sealing.
This phenomenon, however, is not necessarily a defect of the shock absorber.
Please be careful, though, as a faulty seal would cause an oil leak that shows a similar appearance.

Guideline for Judgment by Visual Inspection.

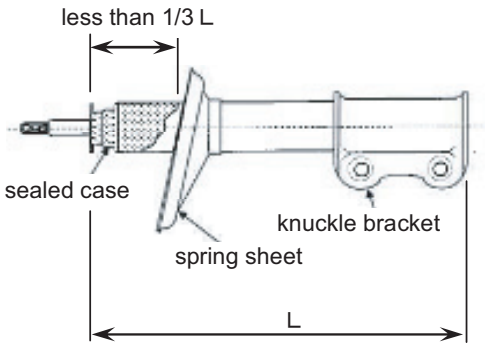
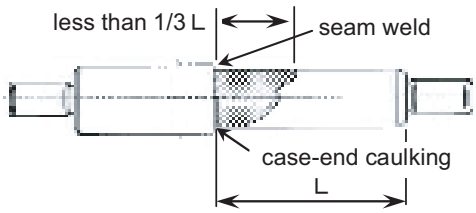
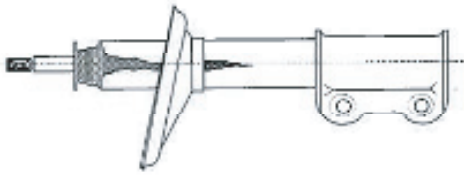
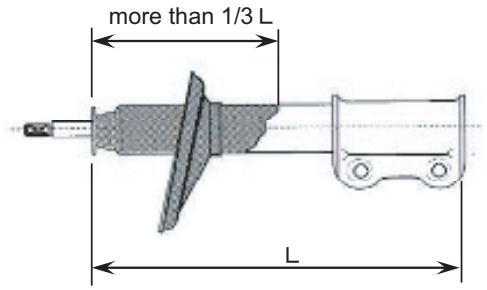
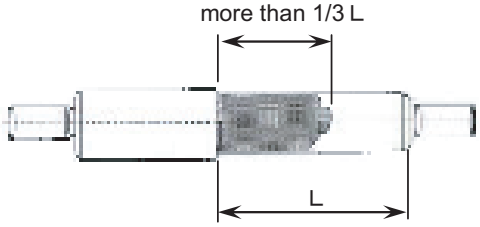
A diagnostic flowchart has been established along with guidelines to help perform shock absorber oil leak judgment. When inspecting shock absorbers, follow the flowchart below and if there is a visible oil adhesion, determine the level of severity, refer to the chart on the next page, then act accordingly.

Level 1: normal oil adhesion, no defect - no action required.
Levels 2 and 3: oil leak due to defect - action required



Attachment 1 (2/2)

Guideline for Determining Oil Leak Severity.

	Shock absorber with spring sheet	Shock absorber with out spring shee	Remarks
Level 1			Trace of oil is less than 1/3 of case length **This level of oil trace pattern does not suggest a progressive oil leak.** **Oil trace may appear dry due to adhession of dirt and dust.**
Level 2			Narrow width oil drippage pattern (length irrelevant) **Trace of oil of this pattern suggests an oil leak over a short period of time.**
Level 3			Trace of oil is more than 1/3 of case length **Trace of oil of this level indicates an oil leak.**